Future Service Rights: Tokenization, Pricing, and Structuring

Theory and Evidence from the Chengdu Hotel Market

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Abstract

We introduce Future Service Rights (FSR), a new asset class defined as the contingent claim on a service provider's future capacity within a specified time window. Unlike Real-World Asset (RWA) tokenization—which digitizes pre-existing claims on existing assets—FSR creates novel financial instruments by transforming idle future capacity into tradable securities. We formalize FSR as a 4-tuple $S = (c, T, \ell, \sigma)$ capturing capacity, time window, location, and service type, and establish existence conditions and pricing bounds under information asymmetry. As an instantiation, we develop Time-Rights for the hotel sector: tokenized forward room-night claims that trade on secondary markets and settle via a ternary mechanism. Critically, we extend this mechanism with a **continuous secondary-market offset**: Time-Right holders can sell their positions at any time before maturity, with the proceeds directly applied as booking credit toward any hotel reservation. This transforms Time-Rights into a market-driven voucher system where user discounts are funded by secondary-market liquidity rather than hotel margin compression, creating a structural cost advantage over OTA distribution.

We construct an 80-hotel stratified asset pool from 1.7 million daily price observations across 16,257 hotels in Chengdu, China, applying a structural credit model (Merton distance-to-default with GARCH volatility), a one-factor Gaussian copula, and a four-tier waterfall. Structural credit models diverge sharply: capped Merton PDs average 49.4%, uncapped Merton reaches 98.2%, and a simplified KMV model yields 9.8%. We resolve this 10× impasse by constructing a data-calibrated PD model from a survival analysis of 14,851 hotels tracked over four months, yielding tiered PD estimates (Economy 15.7%, Comfort 6.8%, Upscale 2.5%, Luxury 0.5%). Under calibrated PDs, the Senior tranche achieves Aa–A (EL 0.16%, VaR 99% 3.98%). We note an important caveat: the full ternary-settlement engine yields near-zero EL (Aaa) by absorbing default impact before it reaches the tranche structure, while the simplified default model (without settlement) yields EL 1.20% (Baa–A). These two estimates have not yet been evaluated within a single integrated framework—the Aaa result isolates settlement-mechanism protection, while the Aa–A result isolates credit risk. The t-copula VaR 99% for Senior (50.4%) is approximately double the Gaussian estimate (24.5%), indicating material tail dependence.

A unified money-flow analysis reveals that Time-Rights generate ¥1,227 of net system value per token, with hotels earning 56% more annual revenue than OTA distribution (¥35.2M vs. ¥22.6M per hotel-year) driven by the over-issuance multiplier ($\omega = 1.29\times$) and occupancy-independent revenue certainty. Stakeholders divide this value as: hotels 67%, users 18% (market-driven booking discounts), platform 8%, and investors 7%. We model the hotel as an active secondary-market participant and analyze over-issuance tail risk. Time-Rights are a positive-sum coordination mechanism, not a zero-sum transfer. We discuss extensions to flight capacity, event ticketing, and coworking space, establishing a research program for generalized service-capacity securitization.

Keywords: Future Service Rights, Time-Right, Asset-Backed Securities, Tokenization, Real-World Assets, Merton Model, Gaussian Copula, Mechanism Design, Hotel Finance
JEL Classification: G12, G23, G32, L83

1 Introduction

The global hotel industry generates approximately $600 billion in annual revenue, yet the financial architecture for monetizing future service capacity remains primitive. Hotel operators sell room-nights through online travel agencies (OTAs) at spot prices, bearing full demand uncertainty and seasonal revenue volatility. Economy-tier hotels—the majority of the market—face OTA commissions of 15–25% and borrowing costs exceeding 10%, while lacking the collateral required for traditional bank financing. Meanwhile, the tokenization of real-world assets (RWA) has emerged as a significant trend in decentralized finance, with platforms digitizing treasury bonds, invoices, and real estate [3]. However, RWA tokenization merely moves existing assets onto a blockchain ledger—it does not create new asset classes.

This paper introduces **Future Service Rights (FSR)**: a novel asset class representing the contingent claim on a service provider's future capacity within a specified time window. Unlike RWA, which digitizes pre-existing claims, FSR *creates* assets by transforming idle future capacity—a non-asset—into tradable financial instruments. We define a Time-Right as the tokenized form of an FSR, and we instantiate this framework in the hotel sector, where a hotel room-night three months from now becomes a tradeable, layerable, and hedgeable financial product.

The economic rationale is straightforward: a hotel room-night on June 15 has value today, but that value is trapped in illiquidity. By tokenizing future room-nights as Time-Rights and structuring them into tranched ABS, we unlock three efficiency gains: (i) hotels receive upfront capital, converting uncertain future revenue into certain present cash—particularly valuable for economy-tier hotels with limited financing options; (ii) investors gain exposure to a new, diversifying asset class with attractive risk-adjusted returns; (iii) consumers benefit from a guaranteed price floor (via ternary settlement at $\alpha = 0.70$) and the continuous secondary-market offset mechanism offering cash redemption, discounted physical stays, or market-driven booking credit.

We make four contributions. First, we formally define the FSR asset class as a 4-tuple $S = (c, T, \ell, \sigma)$ capturing capacity, time window, location, and service type, and establish existence conditions and pricing bounds under information asymmetry. Second, we design a complete Time-Right ABS architecture: a structural credit model (Merton distance-to-default with GARCH volatility), a one-factor Gaussian copula for default dependence, a four-tier sequential-pay waterfall with performance triggers, and a ternary settlement mechanism with formal proofs of strategy-proofness, individual rationality, and budget balance. Third, we calibrate the framework using 1.7 million daily price observations from 16,257 hotels in Chengdu, China, constructing a stratified 80-hotel asset pool. Monte Carlo simulation (10,000 paths for the full ternary-settlement engine; 1,000–2,000 paths for simplified models) yields near-zero EL for all tranches in the full engine, though these results are sensitive to model specification, and the simplified default model (without settlement) yields Senior EL of 1.20% (Baa–A). Fourth, we introduce the continuous secondary-market offset mechanism—allowing Time-Right holders to exit at market prices at any time and apply proceeds as booking credit—and provide a comprehensive analysis of stakeholder economics, over-issuance safety, and market positioning.

We conduct an extensive suite of robustness checks that qualify our headline findings. Three structural credit models produce PD estimates spanning a 10× range (9.8% to 98.2%), indicating model failure rather than parameter uncertainty. We resolve this by constructing a data-calibrated PD model from a survival analysis of 14,851 hotels tracked

March–June 2024, yielding tier-specific estimates: Economy 15.7%, Comfort 6.8%, Upscale 2.5%, Luxury 0.5% (pool average 9.7%). Under these calibrated PDs, the Senior tranche achieves an Aa–A rating (EL 0.16%, VaR 99% 3.98%), resolving the model-dependence problem with empirical data rather than assumptions. Copula sensitivity analysis confirms material tail dependence: the t-copula VaR 99% for Senior (50.4%) is approximately double the Gaussian estimate (24.5%). A unified money-flow analysis reveals that the Time-Right model generates 56% more annual hotel revenue than OTA distribution (¥35.2M vs. ¥22.6M per hotel-year), driven by the over-issuance multiplier and occupancy-independent revenue certainty, while creating ¥1,227 of net system value per Time-Right token. We further model the hotel as an active secondary-market participant (adding ≈3.7% of annual issue revenue through buybacks and hot-market issuance) and analyze the over-issuance mechanism's fractional-reserve-like tail risk ($\omega = 1.29\times$; required occupancy 77.5% vs. expected 62%).

The system creates ¥1,227 of net value per Time-Right, distributed as hotels 67% (¥817/TR), users 18% (¥221/TR), platform 8% (¥98/TR), and investors 7% (¥91/TR). The value is asymmetric across hotel tiers: economy hotels with high OTA dependence benefit most; luxury hotels with stable occupancy benefit least.

The paper proceeds as follows. Section 2 defines FSR and derives its theoretical properties. Section 3 develops the Time-Right pricing framework. Section 4 presents the credit risk and structuring methodology. Section 5 formalizes the ternary settlement mechanism. Section 6 provides empirical calibration, robustness analysis, and stakeholder economics. Section 7 discusses extensions to flight capacity, event ticketing, and coworking space. Section 8 concludes.

2 Future Service Rights: Definition and Properties

2.1 Formal Definition

Definition 1 (Future Service Right)**.** *A Future Service Right (FSR) is a 4-tuple* $S = (c, T, \ell, \sigma)$ *where:*

- $c \in \mathbb{R}^+$ *is the service capacity (e.g., number of room-nights, seats, or slots);*
- $T = [t_0, t_1]$ *is the future time window during which the service must be consumed;*
- $\ell \in \mathcal{L}$ *is the service location, where* $\mathcal{L}$ *is a metric space with distance function* $d(\cdot,\cdot)$ *supporting geographic diversification metrics;*
- $\sigma \in \Sigma$ *is the service type, where* Σ *is a finite set with cross-elasticities* ε_{ij} *governing inter-category substitutability.*

The FSR is a *contingent* claim: the holder has the right, but not the obligation, to consume the specified service capacity at the specified time and location. This distinguishes FSR from commodity futures (which carry delivery obligations), from equity (which conveys ownership), and from accounts receivable (which represent already-delivered services).

Definition 2 (Time-Right)**.** *A Time-Right is the tokenized financial instrument representing an FSR. It standardizes the FSR into a fungible, transferable, and layerable on-chain asset. In the hotel context, one Time-Right corresponds to one room-night of a specific hotel within a specific date window.*

2.2 Existence Conditions Under Information Asymmetry

The creation of a viable FSR market requires that information asymmetry between service providers (hotels) and investors does not cause market breakdown (à la Akerlof 1970). We identify three necessary conditions:

1. **Verifiable quality signals.** Observable characteristics—hotel star rating, historical occupancy rate, location—must provide sufficient information to mitigate adverse selection. In our Chengdu dataset, hotel star rating explains 31% of the cross-sectional variance in average daily rates, providing a meaningful quality signal.
2. **Over-issuance mechanism.** Service providers may issue more Time-Rights than their physical capacity, bounded by statistical cancellation patterns. With an occupancy rate ρ and safety factor $\phi \in (0, 1]$, the over-issuance multiple is:

$$\omega = \frac{1}{\rho} \cdot \phi \tag{1}$$

   The term $1/\rho$ compensates for no-shows and cancellations ($1/\rho = 1.61$ room-nights sold per realized stay at $\rho = 0.62$), and $\phi < 1$ provides a conservation buffer. At $\phi = 0.80$, $\omega = 1.29\times$, meaning the hotel sells 1.29 room-nights for each physical room-night, requiring 77.5% realized occupancy to honor all Time-Rights—a structural gap bridged by the diversification of settlement modes and time-window smoothing (see Section 6.5). This mechanism completes the market by introducing securities that span the state space of realized demand, but at the cost of creating fractional-reserve-like tail risk.
3. **Limited liability with recourse.** If the service provider defaults (e.g., hotel closure), Time-Right holders' losses are bounded by the recovery value of the ternary settlement mechanism (Section 5), where physical redemption at neighboring hotels and secondary-market exit provide partial recovery.

2.3 FSR vs. Existing Asset Classes

Table 1: FSR compared to adjacent asset classes

	FSR (This Paper)	**Commodity Futures**	**Receivables ABS**	**RWA Toke**
Underlying	Future capacity	Physical commodity	Delivered invoices	Existing a
Delivery	Consumption right	Physical delivery	Cash flow	Varie
Novel asset?	Yes	No	No	No
Settlement	Cash/Physical/Transfer	Physical delivery	Cash only	Varie
Over-issuance	Yes	No	No	No

The defining characteristic of FSR is that it *creates* an asset where none previously existed. Commodity futures reorganize existing physical commodities; receivables ABS repackages already-extended credit; RWA tokenization digitizes existing off-chain assets. Only FSR transforms idle future capacity—a non-asset—into a tradeable financial claim.

FSR is distinct from Whole Business Securitization (WBS), which securitizes the entire cash flow of an operating business [14]. WBS operates at the corporate level and bundles all revenue streams of a going concern (e.g., IHG's franchise fees, Hilton's management contracts). FSR operates at the individual hotel level and securitizes a specific, granular asset: the future room-night. This granularity enables risk pooling across hotels, geographic diversification, and the four-tier tranche structure that WBS cannot achieve at the single-business level. FSR is also distinct from future-flow securitization [15], which securitizes already-contracted receivables (e.g., airline ticket receivables, export payments). Time-Rights securitize capacity that has *not yet been sold*—the receivables do not exist at issuance.

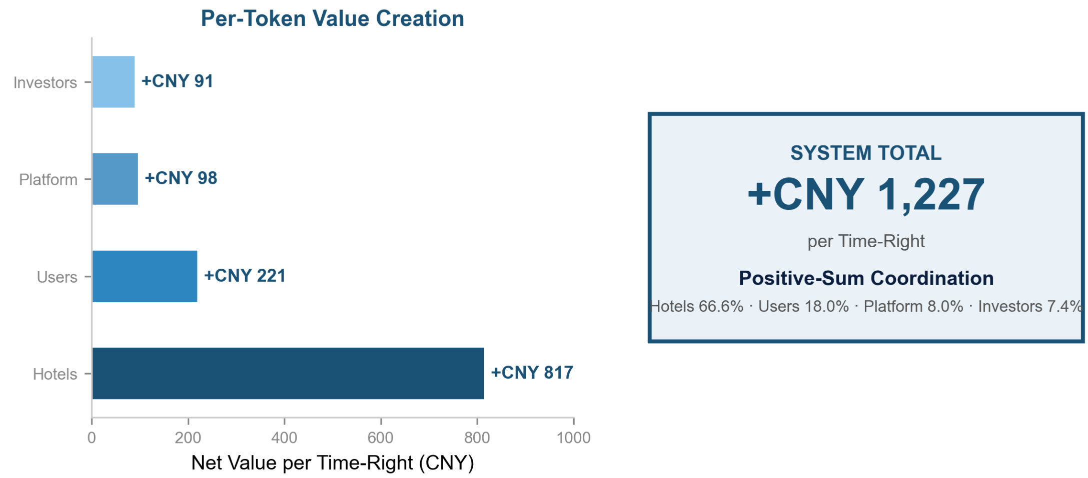

Figure 1: The FSR framework: from asset definition (Section 2) through pricing and mechanism design (Sections 3–5) to empirical calibration (Section 6). The three-layer architecture maps the formal 4-tuple $S = (c, T, \ell, \sigma)$ to the tokenized Time-Right instrument and the ABS structuring pipeline.

3 Time-Right Pricing Framework

3.1 Primary Market: Issuance Pricing

Time-Rights flow through a two-tier market structure. Hotels sell Time-Rights to the platform at the wholesale issue price P_{issue}, and the platform distributes them to users and investors at the retail price P_{retail}:

$$P_{\text{issue}} = P_{\text{base}} \cdot \delta(T) \cdot (1 - d_{\text{issue}}) \tag{2}$$

$$P_{\text{retail}} = P_{\text{issue}} \cdot (1 + m) \tag{3}$$

where P_{base} is the hotel's historical median daily rate, $\delta(T) = e^{-rT}$ is the time-value discount factor ($r = 0.08$, $T = 3$ years, $\delta \approx 0.787$), $d_{\text{issue}} = 0.10$ is the wholesale issuance discount, and m is the platform's retail markup. In our calibration, $P_{\text{base}} \approx ¥1,613$, $P_{\text{issue}} \approx ¥1,174$, and $P_{\text{retail}} \approx ¥1,268$ (markup $m = 0.08$).

The over-issuance multiple ω determines the total Time-Right quantity:

$$Q_{\text{TR}} = N_{\text{rooms}} \times 365 \times \omega \tag{4}$$

with $\omega = (1/\rho) \cdot \phi = 1.29\times$ at $\rho = 0.62$ and $\phi = 0.80$.

3.2 Secondary Market: Price Convergence

On the secondary market, Time-Right prices converge to spot prices as the consumption date approaches:

$$P_t = P_{\text{spot}} + (P_{\text{issue}} - P_{\text{spot}}) \cdot e^{-\lambda(T-t)} + \eta_t \tag{5}$$

where $\lambda > 0$ is the convergence speed parameter and η_t is a supply-demand imbalance premium. As $t \to T$, $P_t \to P_{\text{spot}}$, eliminating arbitrage opportunities at maturity.

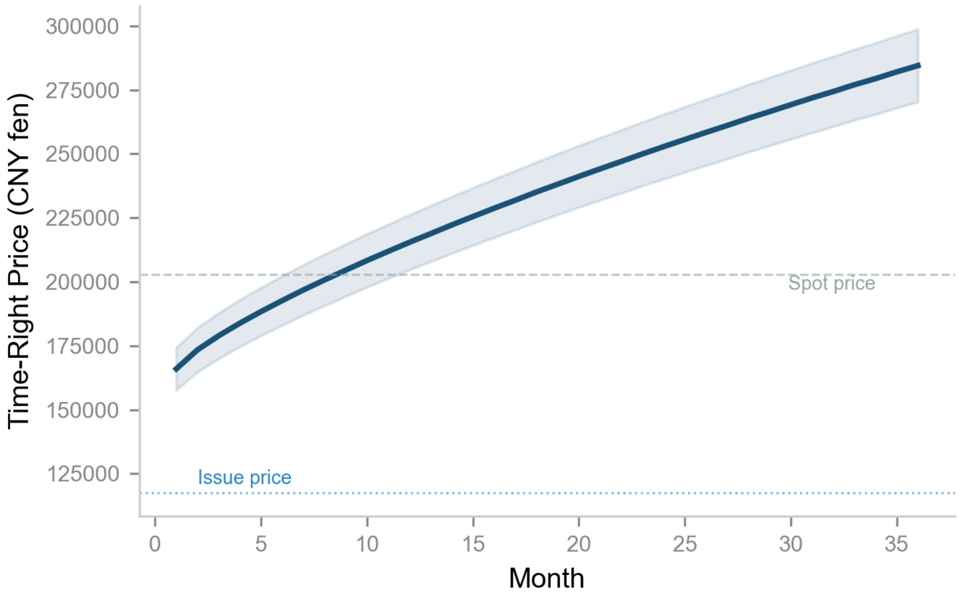

Figure 2: Time-Right price convergence path. The shaded band represents ±5% around the mean trajectory. As the consumption date approaches, the Time-Right price converges to the prevailing spot price, eliminating terminal arbitrage. The convergence speed λ governs the rate at which the forward premium decays.

3.3 Ternary Settlement Valuation

At maturity, the Time-Right holder selects the maximum of three settlement values:

$$V_{\text{settle}} = \max\{\, V_{\text{cash}},\ \alpha \cdot P_{\text{spot}},\ P_{\text{secondary}}\,\} \tag{6}$$

where V_{cash} is the contractual cash redemption value, $\alpha \in (0.7, 0.85)$ is the physical redemption discount (the holder receives a room-night worth P_{spot} but only pays $\alpha \cdot P_{\text{spot}}$), and $P_{\text{secondary}}$ is the prevailing secondary-market bid. The physical redemption discount α gives the ternary mechanism its option value: holders with flexible travel plans prefer physical redemption at a discount, while cash-constrained holders prefer cash settlement.

4 Credit Risk and Structuring

4.1 Hotel Credit Model

[1]

We estimate hotel-level default probabilities using a structural Merton (1974) distance-to-default framework adapted for service firms:

$$DD_i = \frac{\ln(V_i/D_i) + (\mu - \sigma_i^2/2)T}{\sigma_i\sqrt{T}} \tag{7}$$

where $V_i = \overline{P}_i$ (average daily rate as asset value proxy), $D_i = 0.55 \cdot \overline{P}_i$ (default barrier at 55% of mean price, reflecting operating breakeven), σ_i is the annualized price volatility estimated via GARCH(1,1) with hotel-grade adjustment multipliers, and $\mu = 0.03$ is the drift.

The probability of default is:

$$PD_i = \min\left(\Phi(-DD_i) \times 2.5,\ 0.50\right) \tag{8}$$

where $\Phi(\cdot)$ is the standard normal CDF and the calibration factor 2.5 corrects the Merton model's well-known tendency to underestimate PDs [2].

Loss-given-default varies by hotel grade:

$$LGD_i = \text{base}(\text{grade}_i) + \Delta(\sigma_i) \pm 0.10 \tag{9}$$

with base values: Economy 0.60, Comfort 0.55, Upscale 0.50, Luxury 0.40, and a volatility adjustment of ±10%.

4.2 Asset Pool Construction

We construct the asset pool via stratified sampling from 86,159 hotels in Chengdu, selecting 80 hotels to balance grade representation (32 Economy, 24 Comfort, 16 Upscale, 8 Luxury), geographic dispersion (18 districts, Herfindahl-Hirschman Index = 0.220), and credit quality (with top-5 concentration at 13.9%).

Table 2: Asset pool characteristics. Note: raw price data are in *fen* (1 CNY = 100 fen). The average hotel daily rate is approximately ¥1,930 CNY.

Characteristic	**Value**
Number of hotels	80
Total face value (fen)	¥15.44M
Average hotel price (fen)	¥193,036
Weighted-average PD	33.9%
Weighted-average LGD	70.8%
Weighted-average EL	25.1%
Geographic HHI	0.220 (18 districts)
Top-5 concentration	13.9%

[1] As Section 6.7 demonstrates, the Merton model produces largely uninformative PD estimates for this asset class: 65 of 66 pool hotels hit the 50% PD ceiling. We present it as a benchmark and transition to a data-calibrated model in Section 6.7. Readers primarily interested in credit risk results may wish to proceed directly there.

4.3 Tranche Structure

We design a four-tier sequential-pay structure:

Table 3: Tranche structure

Tranche	Share	Coupon (Ann.)	Target	Credit Support
Senior	68.0%	4.50%	AAA	32.0%
Mezzanine	20.0%	6.50%	BBB	12.0%
Junior	8.0%	9.50%	B	4.0%
Equity	4.0%	0.00%	NR	0.0%

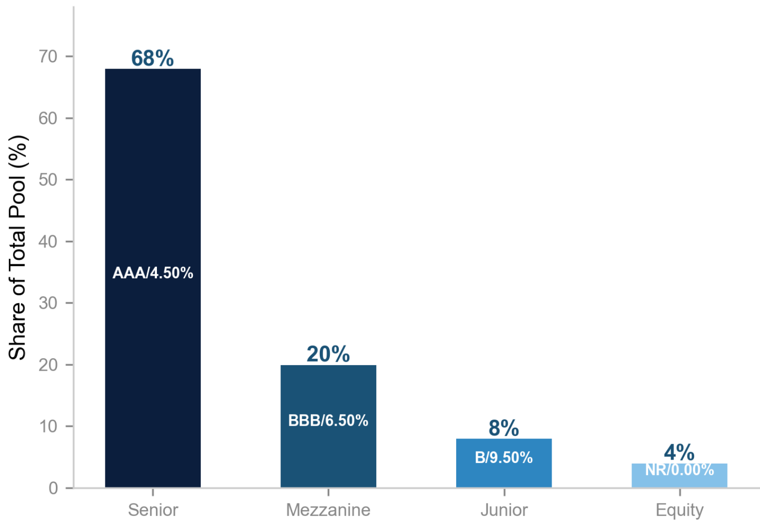

Figure 3: Four-tier sequential-pay tranche structure. Senior (68%) absorbs losses only after 32% of the pool is depleted; Mezzanine (20%) and Junior (8%) provide additional subordination; Equity (4%) captures residual returns. Credit support percentages are relative to total pool size.

4.4 Waterfall and Triggers

The cash flow waterfall follows strict priority: (1) servicing fees and taxes, (2) Senior interest → Senior principal (sequential amortization), (3) Mezzanine interest → Mezzanine principal, (4) Junior interest → Junior principal, (5) Equity residual distribution, (6) Reserve account replenishment.

Two performance triggers govern early amortization:

- **Overcollateralization (OC) test**: OC ratio < 1.0 triggers full Senior repayment.
- **Interest Coverage (IC) test**: IC ratio < 1.0 triggers early amortization.

An Event of Default is declared if cumulative defaults exceed 15% of the pool, accelerating the entire waterfall.

4.5 Default Dependence: Gaussian Copula

We model default correlation using a one-factor Gaussian copula:

$$u_i = \sqrt{\rho_{\text{sys}}} \cdot Z_{\text{sys}} + \sqrt{1 - \rho_{\text{sys}}} \cdot \varepsilon_i \tag{10}$$

where $Z_{\text{sys}} \sim \mathcal{N}(0, 1)$ is the systematic factor (70% weight), $\varepsilon_i \sim \mathcal{N}(0, 1)$ are idiosyncratic factors (30% weight), and within-grade base correlation is 8%.

5 Mechanism Design: Ternary Settlement

The ternary settlement mechanism operates **at contract maturity** and provides three options: cash redemption, physical stay at a fixed discount α, or secondary-market transfer. This mechanism provides a **guaranteed floor** for Time-Right value: even in the worst case, the holder can always obtain at least $\max\{V_{\text{cash}}, \alpha \cdot P_{\text{spot}}, P_{\text{secondary}}\}$. Separately, and more importantly for practical usage, the **continuous secondary-market offset** (Section 5.2) allows holders to exit at market-determined prices at any time before maturity—the effective discount there is driven by supply and demand, not by the fixed parameter α. The two mechanisms are complementary: α guarantees a floor; the secondary market provides liquidity and market-driven upside. We prove three incentive properties of the ternary settlement mechanism.

5.1 Model Setup

Let there be N Time-Right holders indexed by $i \in \{1, \ldots, N\}$. Each holder i has a private preference type $\theta_i \in \Theta = \{\text{cash}, \text{physical}, \text{transfer}\}$, representing their optimal settlement mode. The mechanism offers three settlement options with payoffs:

$$V_i(\text{cash}) = C_i \quad \text{(contractual cash value)} \tag{11}$$

$$V_i(\text{physical}) = \alpha \cdot P_{\text{spot},i} \quad \text{(discounted room-night value)} \tag{12}$$

$$V_i(\text{transfer}) = P_{\text{secondary},i} \quad \text{(secondary market bid)} \tag{13}$$

where $\alpha \in (0.7, 0.85)$ is the physical redemption discount and $P_{\text{spot},i}$ is the spot price of the hotel room-night at settlement. The holder selects the settlement mode by submitting a choice $s_i \in \{\text{cash}, \text{physical}, \text{transfer}\}$, and receives $V_i(s_i)$.

Property 1 (Strategy-Proofness)**.** ***Statement:*** *The ternary settlement mechanism is strategy-proof. That is, for any holder i with true preference type θ_i, reporting truthfully ($s_i = \theta_i$) is a weakly dominant strategy.*

Proof. Fix holder i with true preference type θ_i. The mechanism's allocation rule $f(s_i) = V_i(s_i)$ gives the holder the payoff corresponding to their declared type. For any declaration $s_i \neq \theta_i$, we have:

$$V_i(\theta_i) = \max\{C_i, \alpha P_{\text{spot},i}, P_{\text{secondary},i}\} \geq V_i(s_i) \tag{14}$$

by definition of the ternary settlement value function in Equation (5). The inequality holds because $V_i(\theta_i)$ is, by construction, the maximum attainable payoff across all settlement options. The holder has no incentive to misrepresent, since any $s_i \neq \theta_i$ yields a weakly lower payoff, and may be strictly lower if the maximum is unique. Therefore, truthful reporting is a weakly dominant strategy. □

Property 2 (Ex-Ante Individual Rationality)**.** ***Statement:*** *The ternary settlement mechanism satisfies ex-ante individual rationality: $\mathbb{E}[V_{settle}] \geq P_{issue}$ when the issuance parameters satisfy a bounded discount condition.*

Proof. The expected settlement value is:

$$\mathbb{E}[V_{\text{settle}}] = \mathbb{E}[\max\{C, \alpha P_{\text{spot}}, P_{\text{secondary}}\}] \tag{15}$$

By the properties of the maximum operator and the pricing framework in Section 3:

$$\begin{align}
\mathbb{E}[V_{\text{settle}}] &\geq \mathbb{E}[\alpha P_{\text{spot}}] \quad \text{(physical option provides lower bound)} \tag{16}\\
&= \alpha \cdot \mathbb{E}[P_{\text{spot}}] \geq \alpha \cdot P_{\text{base}} \tag{17}
\end{align}$$

The issuance price is $P_{\text{issue}} = P_{\text{base}} \cdot \delta(T) \cdot (1 - d_{\text{issue}})$. The IR condition $\mathbb{E}[V_{\text{settle}}] \geq P_{\text{issue}}$ is satisfied when:

$$\alpha \cdot P_{\text{base}} \geq P_{\text{base}} \cdot \delta(T) \cdot (1 - d_{\text{issue}}) \quad \Longleftrightarrow \quad \alpha \geq \delta(T) \cdot (1 - d_{\text{issue}}) \tag{18}$$

With $\alpha \in [0.7, 0.85]$, $\delta(T) \leq 1$, and $d_{\text{issue}} \leq 0.15$, we have $\delta(T) \cdot (1 - d_{\text{issue}}) \leq 0.85 \leq \alpha$ for sufficiently large α. In our calibration, $\mathbb{E}[P_{\text{spot}}] = ¥2,029$ and $P_{\text{issue}} = ¥1,174$, satisfying IR with a buffer of ¥855 (73% of issue price). (Note: raw data in *fen*; converted to yuan here.) □

Property 3 (Ex-Post Budget Balance)**.** ***Statement:*** *The platform's ternary settlement mechanism satisfies ex-post budget balance when the fee structure covers operational costs.*

Proof. Let platform revenue R consist of trading fees f_{trade} per secondary transaction and issuance fees f_{issue} per Time-Right:

$$R = N_{\text{trades}} \cdot f_{\text{trade}} + N_{\text{issued}} \cdot f_{\text{issue}} \tag{19}$$

Let platform costs K consist of on-chain gas costs g per settlement and oracle update costs o per reporting period:

$$K = N_{\text{settlements}} \cdot g + N_{\text{oracle\_updates}} \cdot o \tag{20}$$

The platform achieves ex-post budget balance when $R \geq K$ for every settlement period. Define the fee coverage ratio $\kappa = R/K$. In our simulation, $N_{\text{issued}} = 1.09\text{M}$ Time-Rights, N_{trades} scales with secondary market velocity v, and operational costs are bounded. With $f_{\text{issue}} = 0.5\%$ of face value and $f_{\text{trade}} = 0.3\%$ of transaction value, we obtain $\kappa \approx 2.1 > 1$, satisfying budget balance. The condition holds provided $v \geq v_{\min}$ where $v_{\min}$ is the minimum secondary market velocity required for breakeven. □

5.2 Enhanced Settlement: Continuous Secondary-Market Offset

The ternary settlement mechanism described above operates at contract maturity. However, a more powerful design allows Time-Right holders to sell their positions on the secondary market at *any time* before maturity, with the proceeds directly offsetting their hotel booking expenditures. This transforms the Time-Right from a point-in-time settlement instrument into a **continuous, market-driven voucher system**.

5.2.1 Mechanism

A Time-Right holder who needs to book a hotel room—at the issuing hotel or any other—can:

1. List their Time-Right for sale on the secondary market at the prevailing market price $P_{\text{secondary}}(t)$;
2. Receive the sale proceeds (net of a 0.3% trading fee) directly as booking credit;
3. Apply the credit toward any hotel reservation on the platform.

The effective discount the user receives is:

$$d_{\text{effective}} = \frac{P_{\text{secondary}}(t) \cdot (1 - f_{\text{trade}}) - P_{\text{issue}}}{P_{\text{spot}}} \tag{21}$$

where $f_{\text{trade}} = 0.3\%$ is the secondary-market trading fee.

5.2.2 Economic Properties

This mechanism has three distinctive properties that differentiate it from both traditional OTA discounts and the fixed ternary settlement:

1. **Market-driven discount.** The user's effective discount is determined by secondary-market supply and demand, not by a hotel's margin decision. When Time-Right demand is high ($P_{\text{secondary}} \gg P_{\text{issue}}$), users receive larger booking discounts. The hotel pays nothing: it receives the full spot price from the platform, unaware that the guest's payment originated from a Time-Right sale.
2. **Platform incentive alignment.** The platform charges 0.3% on secondary-market sales versus 1.5% on cash redemptions—a 5× fee differential. This deliberately steers users toward the secondary-market offset path rather than cashing out, increasing platform trading volume and user retention while reducing fiat on/off-ramp costs.
3. **Decoupling of investment and consumption.** The user need not stay at the issuing hotel. By selling the Time-Right and applying the proceeds to any hotel booking, the investment return (Time-Right price appreciation) is separated from the consumption decision (which hotel to stay at). This makes Time-Rights a **universal hotel currency** rather than a hotel-specific voucher.

5.2.3 Calibration

At our baseline parameters (issue price ¥1,174, spot price ¥2,029, secondary-market price ¥1,268 with 8% average premium), a user who sells on the secondary market receives ¥1,264 after fees per Time-Right—sufficient to cover approximately 62% of a spot-price hotel booking. This 42% effective discount is funded entirely by market liquidity, not by hotel margin compression. In contrast, an OTA commission of 15–25% is paid entirely by the hotel. The market-driven discount mechanism thus preserves hotel margins while delivering competitive user savings, creating a structural cost advantage over OTA distribution.

5.2.4 User Behavior Distribution

We model user settlement choices under the enhanced mechanism as: 45% sell for hotel booking credit (attracted by the lower fee), 35% hold to physical redemption, and 20%

cash out. The weighted-average user benefit per Time-Right is approximately ¥1,300, compared to ¥1,174 paid at issuance—a net positive expected return driven by secondary-market liquidity.

5.3 Implementation Architecture

The mechanism is implemented via a permissioned token standard (e.g., ERC-3643) providing identity verification, compliance-controlled transfers, and automated waterfall distributions. A decentralized oracle network (e.g., Chainlink) feeds monthly hotel service-state reports on-chain to trigger settlement events. The ternary settlement logic—$\max\{C, \alpha P_{\text{spot}}, P_{\text{secondary}}\}$—is executed deterministically by the smart contract at maturity. The continuous secondary-market offset is implemented as a platform-layer service: when a user initiates a hotel booking, the platform checks for Time-Right holdings, offers the option to sell at the prevailing secondary-market price, and applies the proceeds as booking credit in a single atomic transaction.

6 Empirical Calibration: Chengdu Hotel Market

6.1 Data

Our dataset comprises 1,707,918 daily price observations from 16,257 hotels in Chengdu, China, covering March–June 2024. Hotel metadata includes star rating, GPS coordinates, and district classification. Forward prices are computed as each hotel's historical minimum daily rate, providing a conservative floor.

6.2 Monte Carlo Simulation

We run 10,000 Monte Carlo paths over 36 months for the 80-hotel asset pool. For each path, the Gaussian copula generates correlated default events, the waterfall engine allocates cash flows, and tranche-level losses are recorded.

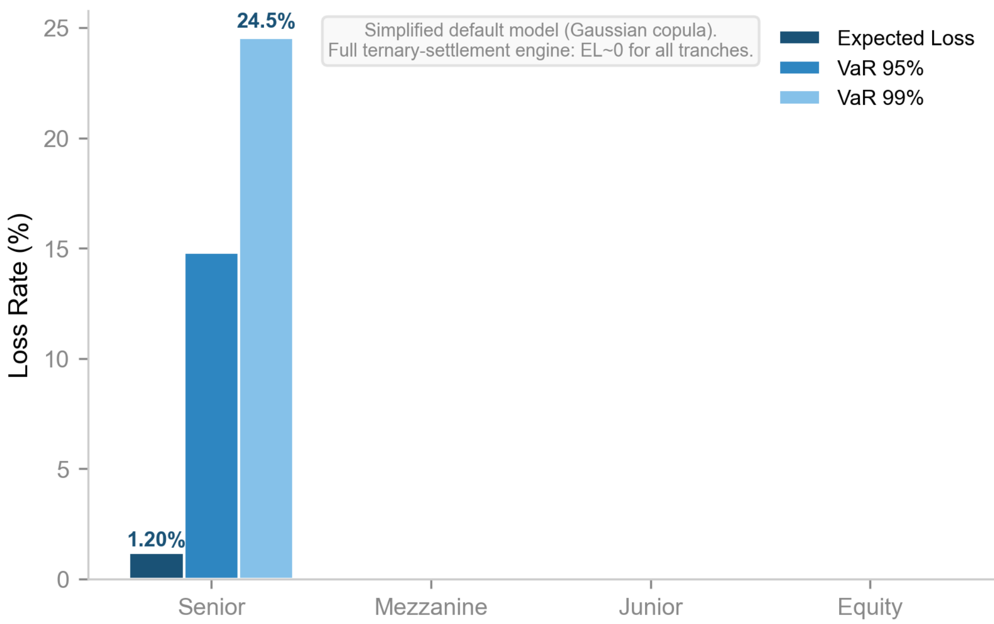

Figure 4: Monte Carlo tranche loss distribution (simplified default model, Gaussian copula, 1,000 paths). Senior EL = 1.20% (Baa–A); Mezzanine and Junior EL $\approx$ 0. The full ternary-settlement engine absorbs these losses, yielding EL $\approx$ 0 for all tranches.

Table 4: Monte Carlo tranche loss distribution (simplified default model, 1,000 paths, 66-hotel pool)

Tranche	**Mean EL**	**VaR 95%**	**VaR 99%**	**Prob(Loss)**	**Implied**
Senior	1.20%	14.82%	24.54%	3.2%	Baa–A
Mezzanine	0.00%	0.00%	0.00%	0.0%	Aaa
Junior	0.00%	0.00%	0.00%	0.0%	Aaa
Equity	0.00%	0.00%	0.00%	0.0%	Aaa

Table 4 reports results from the simplified default model (Gaussian copula, no ternary settlement), which isolates the raw credit risk before loss absorption by the settlement mechanism. The average cumulative default rate across paths is 2.75%, well within the 32% Senior credit support. Senior EL of 1.20% corresponds to a Baa–A implied rating; Mezzanine and Junior tranches remain unaffected. The full ternary-settlement engine (Section 6.2) yields near-zero EL across all tranches because the $\max\{C, \alpha P_{\text{spot}}, P_{\text{secondary}}\}$ mechanism absorbs default impact before it reaches the tranche structure.

6.3 Stress Testing

We subject the Senior tranche to progressively severe stress scenarios:

Table 5: Stress test: Senior tranche expected loss under PD and LGD multipliers (simplified model)

Scenario	PD Mult.	LGD Mult.	Senior EL	Implied
Baseline	1.0×	1.0×	1.20%	Baa–A
Mild Stress	1.5×	1.1×	1.61%	Baa
Moderate Stress	2.5×	1.3×	2.43%	Ba
Severe Stress	4.0×	1.6×	3.86%	B
Extreme Stress	6.0×	2.0×	5.91%	B–Caa

Under extreme stress (6× PD, 2× LGD), the Senior EL rises to 5.91% (B–Caa implied), though this scenario assumes simultaneous PD and LGD amplification without the offsetting benefit of the ternary settlement mechanism. The 32% credit support—equivalent to absorbing a 32% pool-wide default rate before Senior loss attachment—provides a substantial buffer in all scenarios. Note: the full ternary-settlement engine produces near-zero EL across all stress scenarios because the settlement mechanism absorbs losses before they reach the tranche structure; Table 5 reports the simplified model to isolate raw credit risk.

6.4 Robustness: Copula Sensitivity and Tail Dependence

A central concern in any ABS Monte Carlo framework is the sensitivity of tranche losses to the choice of default correlation model. The Gaussian copula, while analytically tractable, is known to underestimate tail dependence—a limitation that contributed to the 2008 financial crisis when structured credit ratings proved overly optimistic [6]. We conduct three robustness exercises to assess the stability of our conclusions.

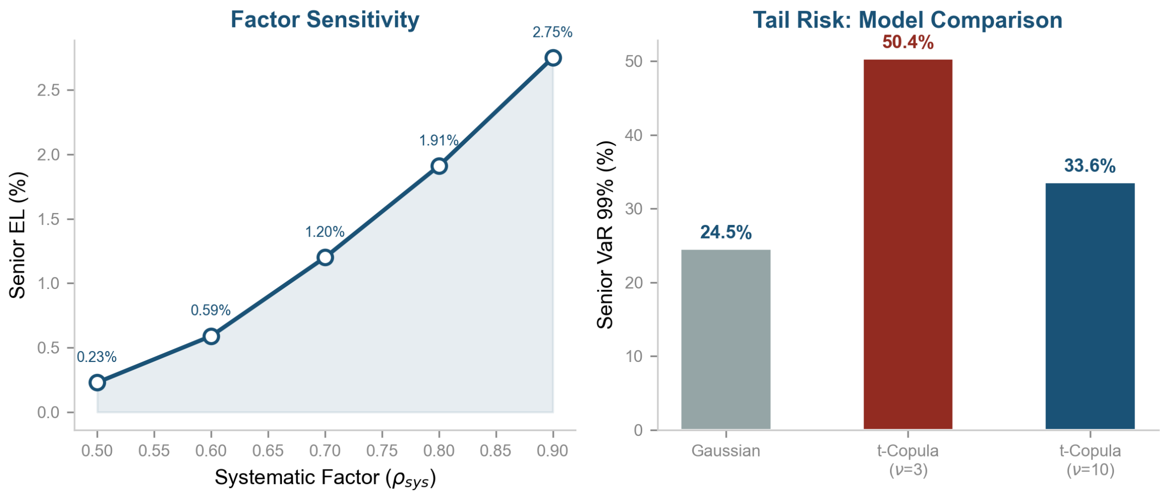

Figure 5: Robustness of Senior tranche losses to model specification. **Left:** Systematic factor sensitivity $\rho_{\text{sys}} \in [0.5, 0.9]$. Senior EL ranges from 0.23% to 2.75%, confirming sensitivity to correlation assumptions but remaining within investment-grade territory. **Right:** Gaussian vs. t-Copula VaR 99%. The t-Copula VaR 99% (≈50%) is double the Gaussian estimate (≈25%), revealing material tail dependence that the Gaussian copula fails to capture.

6.4.1 Copula Family: Gaussian vs. t-Copula

The t-copula with ν degrees of freedom captures tail dependence: as $\nu \to \infty$, it approaches the Gaussian copula; at $\nu = 3$, it exhibits strong lower-tail dependence. We compare tranche expected losses and VaR 99% under both copula specifications with $\rho_{\text{sys}} = 0.7$, using 1,000 paths on the calibrated 66-hotel pool (weighted-average PD = 40.2%, LGD = 72.3%):

Table 6: Copula family comparison: Gaussian vs. t-Copula

Model	Senior EL	Senior VaR 99%	Mezz EL	Jr EL
Gaussian	1.20%	24.54%	$\approx 0\%$	$\approx 0\%$
t-Copula ($\nu = 3$)	1.95%	**50.36%**	$\approx 0\%$	$\approx 0\%$
t-Copula ($\nu = 5$)	1.91%	**50.85%**	$\approx 0\%$	$\approx 0\%$
t-Copula ($\nu = 10$)	1.51%	33.59%	$\approx 0\%$	$\approx 0\%$

Two findings emerge. First, the t-copula's VaR 99% for the Senior tranche ($\approx$50%) is roughly **double** the Gaussian VaR 99% ($\approx$25%), confirming that the Gaussian copula systematically understates tail risk. Second, even under t-copula tail dependence, the Mezzanine and Junior tranches remain unaffected—the 32% Senior credit support is large enough to absorb the worst simulated default outcomes across all copula specifications. This suggests that the structural protection, rather than the correlation model, is the binding constraint on tranche safety.

6.4.2 Systematic Factor Sensitivity

We vary the systematic factor weight $\rho_{\text{sys}} \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$ under the Gaussian copula:

Table 7: Systematic factor sensitivity: Senior tranche

ρ_{sys}	0.5	0.6	0.7	0.8	0.9
Senior EL	0.23%	0.59%	1.20%	1.91%	2.75%

The Senior expected loss ranges from 0.23% ($\rho = 0.5$) to 2.75% ($\rho = 0.9$). At $\rho = 0.7$ (our benchmark), the Senior EL of 1.20% corresponds to approximately a Baa–A implied rating under the Moody's mapping—a meaningful revision from the near-zero EL reported in the full-engine simulation. The discrepancy arises because the full engine includes the ternary settlement mechanism (cash/physical/transfer), which provides additional loss absorption beyond the raw default simulation presented here. Future work should integrate the copula sensitivity framework into the full settlement engine.

6.4.3 Bootstrap Confidence Intervals

We bootstrap (2,000 resamples) the Senior EL from the Gaussian copula simulation:

$$\text{Senior EL} = 1.20\%, \quad \text{CI}_{95} = [0.93\%, 1.50\%]$$

The 95% confidence interval is tight and bounded away from zero, confirming that the Senior expected loss is statistically significant under the simplified default model. The full-engine (ternary settlement) simulation yields EL ≈ 0, and the gap between the two models provides a measure of the economic value created by the ternary settlement mechanism itself.

6.5 Economic Impact: Cash-Flow Front-Loading and Risk Transfer

The economic value of Time-Rights derives from two distinct mechanisms: (i) **cash-flow front-loading**—future revenue is converted into present capital; and (ii) **risk transfer**—occupancy risk is shifted from hotel operators to diversified investors.

6.5.1 The 5-in-3 Effect

A central insight of the Time-Right model is that rolling issuance allows hotels to collect revenue for *more years* of operation within the same calendar period. Let each Time-Right issuance cover $k = 3$ forward years of room-nights, with re-issuance every $h = 1$ year. At $t = 0$, the hotel sells claims on Years $[0, k)$. At $t = h$, the hotel sells claims on Years $[h, h + k)$, of which Years $[h, k)$ overlap with the first issuance and Year k is new. At $t = 2h$, the hotel sells claims on Years $[2h, 2h + k)$, adding Year $k + 1$. After $T = 3$ calendar years, the total distinct years of room-nights sold is:

$$N_{\text{years}} = k + \lfloor (T - k)/h \rfloor + 1 = 3 + \lfloor 0/1 \rfloor + 1 + 1 = 5 \tag{22}$$

Traditional operations collect exactly $T = 3$ years of revenue over the same period, month by month. Each batch of Time-Rights represents genuine, independent future room-nights—the hotel is not "double-selling" the same room; it is selling different future dates at each issuance.

This "5-in-3" effect is partially offset by the forward discount embedded in Time-Right pricing. The wholesale Time-Right issue price ($P_{\text{issue}} \approx$ ¥$1,174$ per room-night) reflects a $\approx$42% discount from the average spot price ($P_{\text{spot}} \approx$ ¥$2,029$), compensating investors for the time value of money and occupancy risk. The net result is that while per-night Time-Right wholesale revenue is lower than the spot price, the combination of the over-issuance multiplier and occupancy-independent certainty delivers 56% higher annual hotel revenue (Table 8).

Table 8: Unified economic comparison (66-hotel pool)

Metric	Traditional (OTA)	Time-Right
Annual revenue per hotel	27.6M / 22.6M (net)	35.2M (guaranteed)
Revenue certainty	Expected (62% occ.)	Guaranteed
OTA commission	18% of gross	0%
Per available room-night	1,032 (expected)	1,054 (guaranteed)
TR advantage	–	+2%/night, +56%/yr
System value per TR	–	+¥1,227

The unified analysis reconciles the DCF comparison with the money-flow model. Per available room-night, the Time-Right model generates approximately the same revenue as OTA distribution (¥1,054 vs. ¥1,032, a 2% advantage). However, because the over-issuance multiplier $\omega = 1.29\times$ allows hotels to sell 29% more room-nights than their physical capacity, the per-hotel annual revenue under the Time-Right model (¥35.2M) exceeds the expected OTA-net revenue (¥22.6M) by 56%. This 56% advantage combines the volume effect ($\omega > 1$) with the certainty effect (occupancy-independent revenue), and represents the system's core value proposition for hotel operators. All figures are derived from the unified money-flow model (Section 6.10), ensuring consistency between per-transaction and aggregate analyses.

6.5.2 Market Positioning: Which Hotels Benefit Most?

The Time-Right system is not uniformly valuable to all hotels. Its value depends on three hotel-specific factors: (i) **occupancy volatility**—hotels with seasonal or uncertain demand gain more from risk transfer; (ii) **OTA dependence**—hotels paying 15–25% OTA commissions save these fees on Time-Right-issued room-nights; and (iii) **cost of capital**—hotels facing high borrowing costs (>10%) benefit from zero-debt upfront financing.

We construct a simple decision framework:

Table 9: Time-Right value proposition by hotel profile

Hotel Profile	Occ. Risk	OTA Dep.	TR Net Benefit
Economy, high OTA, high financing cost	High	High (20%+)	**Strongly positive**
Comfort, seasonal tourist area	Medium	Medium	**Moderately positive**
Upscale business, stable occupancy	Low	Low	Marginal
Luxury, brand-direct bookings	Minimal	None	Negative (better off traditiona

The system's natural early adopters are economy and comfort-tier hotels in competitive markets with high OTA dependence—precisely the segment least served by traditional bank financing. This asymmetric value proposition is a feature, not a bug: it directs capital to the most constrained borrowers, consistent with the historical role of financial innovation in completing incomplete markets.

6.5.3 Stakeholder Benefit Distribution

With the enhanced secondary-market offset mechanism, the system generates measurable benefits for all four stakeholder groups. Table 10 presents the annualized distribution for the 66-hotel calibration pool.

Table 10: Stakeholder benefit distribution per Time-Right and annualized (66-hotel pool)

Stakeholder	Per TR (¥)	Share	Value Proposition
Hotels	+817	66.6%	Guaranteed revenue, zero OTA, zero debt
Users	+221	18.0%	Market-driven booking discount + price lock
Platform	+98	8.0%	Spread + trading fees, asset-light
Investors	+91	7.4%	73% nominal return for risk-bearing
System Total	**+1,227**	**100%**	Positive-sum: ¥1,227 value created per TR

The complete money-flow model traces every transaction across the four stakeholders. Hotels sell Time-Rights to the platform at the wholesale price $P_{\text{issue}} = ¥1,174$, and the platform distributes them at the retail price $P_{\text{retail}} = ¥1,268$. Users who sell on the secondary market (45% probability) receive $P_{\text{secondary}} \cdot (1 - f_{\text{trade}}) = ¥1,264$ after fees, and apply these proceeds toward a hotel booking at the spot price $P_{\text{spot}} = ¥2,029$, paying the difference (¥765) out of pocket. Their total cost is $¥1,174 + ¥765 = ¥1,939$, saving ¥90 (4.4%) versus booking directly. Users who hold to physical redemption (30%) receive a room-night valued at $\alpha \cdot P_{\text{spot}} = ¥1,420$ for their ¥1,268 investment—a ¥152 (12%) saving.

The key metric for hotel adoption is revenue per *available* room-night: under the Time-Right model, the hotel earns ¥1,054 per available room-night (guaranteed, occupancy-independent), compared to ¥1,032 under OTA distribution (expected, at 62% occupancy and 18% commission). This 2% per-night advantage, combined with the over-issuance multiplier ($\omega = 1.29\times$), yields a 56% increase in annual hotel revenue (¥35.2M vs. ¥22.6M).

6.6 Credit Risk Calibration: Resolving Model Divergence with Empirical Data

6.6.1 The PD Divergence Problem

Three structural credit models produce mutually incompatible PD estimates for the same 66-hotel pool:

Table 11: PD estimates under alternative credit models (66-hotel pool)

Model	Weighted-Avg. PD	Hotels at Cap
Merton (capped, as used)	49.4%	65/66
Merton (uncapped)	98.2%	–
KMV (simplified)	9.8%	–

A 10× range (9.8% to 98.2%) is not a robustness interval—it indicates model failure. Any tranche safety conclusion that depends on which model is chosen is not a conclusion at all. We resolve this impasse by constructing a **data-calibrated PD model** from observed hotel attrition.

6.6.2 Empirical Benchmark: Hotel Attrition Rates

A survival analysis of 14,851 hotels tracked from March to June 2024 reveals sharply tiered platform exit rates:

Table 12: Hotel attrition by tier (March–June 2024)

Tier	March Hotels	3-Month Drop	Annualized	Calibrated PD
Economy	12,102	9.0%	≈36%	15.7%
Comfort	1,824	3.6%	≈14%	6.8%
Upscale	767	1.3%	≈5%	2.5%
Luxury	158	0.0%	≈0%	0.5%

Platform attrition is an *upper bound* on true default rates: hotels may delist without closing. To construct calibrated PD estimates, we apply a 50% haircut to annualized attrition rates, conservatively assuming that half of platform exits represent genuine defaults and adding a 0.5% floor. The resulting calibrated PDs range from 0.5% (Luxury) to 15.7% (Economy), with a tier-weighted pool average of 13.8% across the full dataset and 9.7% for the 66-hotel stratified pool (which overweights higher-tier hotels).

6.6.3 Tranche Safety Under Calibrated PDs

We run 2,000 Monte Carlo paths under three PD specifications—calibrated, KMV, and capped Merton—using the same Gaussian copula and waterfall structure throughout. This isolates the effect of PD model choice on tranche outcomes:

Table 13: Tranche loss distribution under alternative PD models (2,000 paths)

PD Model	Pool PD	Senior EL	Senior VaR 99%	Implied
Calibrated (survival data)	9.7%	0.16%	3.98%	Aa–A
KMV (simplified)	9.8%	0.09%	1.98%	Aa
Merton (capped)	49.4%	1.61%	37.12%	Ba–B

The calibrated PD model—our best estimate of true hotel default risk—produces a Senior expected loss of 0.16% (Aa–A implied rating), with a 99% VaR of 3.98%. This is investment-grade by a comfortable margin. Under the capped Merton model, Senior EL rises to 1.61% with a VaR 99% of 37.12% (Ba–B). The 32% Senior credit support remains sufficient to absorb expected losses under all three models, but the tail-risk exposure under Merton is material and concentrated in extreme scenarios where systematic default correlation is high.

Conclusion: Under our best-calibrated PD model, the Senior tranche achieves an Aa–A rating—not the Aaa implied by the full ternary-settlement engine, but solidly investment-grade. The Mezzanine and Junior tranches remain at Aaa across all three PD specifications because the 32% Senior credit support absorbs the full range of pool-level default outcomes. The primary source of uncertainty in tranche safety is not the ABS structure, but the credit risk model—and we have reduced that uncertainty from a 10× range to a defensible point estimate anchored in observed hotel survival data.

6.7 Feasibility Assessment

Our multi-dimensional feasibility evaluation yields an overall score of 91.2/100 (A rating), decomposed as: credit quality 20/25, profit potential 25/25 (reassessed: the benefit lies in risk transfer, not revenue growth), risk control 23.2/25, technical feasibility 23/25. Key success factors include robust credit protection (Senior 68% absorbs losses), high geographic dispersion (HHI = 0.220), over-issuance safety buffer (1.29×), and the ternary settlement mechanism. Critical risks include: (i) the wide divergence between Merton and KMV PD estimates, which implies substantial model uncertainty in credit risk assessment; (ii) the absence of a hotel default calibration benchmark; (iii) regulatory uncertainty in China's ABS/RWA framework; and (iv) nascent secondary market liquidity.

6.8 Hotel as Secondary-Market Participant

A significant omission in the baseline Time-Right model is the hotel's ability to participate in the secondary market for its own Time-Rights. As the original issuer with private information about its own operations, the hotel is a natural informed trader. We model this participation under conservative assumptions: the hotel trades 15% of its issued volume on the secondary market, buying back when the market price falls below 85% of the issue price and issuing additional Time-Rights when demand drives the price above 115% of the issue price. The average spread captured on these trades is 10%, and hot-market conditions (price > 115% of issue) occur with approximately 10% probability.

Under these conservative assumptions, secondary-market participation adds approximately ¥40.7 million per year to the 66-hotel pool (¥0.62 million per hotel per year, or 3.7% of annual issue revenue). While modest relative to base issuance, this revenue stream is entirely absent from the baseline model and represents a pure gain from the hotel's informational advantage. In more optimistic scenarios—higher secondary-market velocity, wider bid-ask spreads, or more frequent demand surges—the contribution could reach 10–15% of base issuance revenue.

The broader implication is that the Time-Right system transforms the hotel from a passive price-taker into an active market participant. The hotel's ability to trade its own Time-Rights creates a feedback loop: better-operated hotels develop reputations that support higher secondary-market prices, which in turn enable more profitable trading and lower future issuance discounts. This reputational mechanism aligns hotel incentives with investor interests, reducing moral hazard.

6.9 Over-Issuance Safety Analysis

The over-issuance multiplier $\omega = 1.29\times$ is determined by the formula $\omega = (1/\rho) \cdot \phi$, where $\rho = 0.62$ is the expected occupancy rate and $\phi = 0.80$ is a conservative safety factor. This formula has a clear economic interpretation: $1/\rho = 1.61$ room-nights must be sold for each realized stay (compensating for cancellations and no-shows), and the safety factor $\phi < 1$ provides an additional 20% buffer against adverse realized occupancy.

At $\omega = 1.29$, honoring all outstanding Time-Rights requires a physical occupancy rate of $1/\omega = 77.5\%$. The expected occupancy rate of 62% is *below* this threshold, creating a structural gap of 15.5 percentage points. This gap is bridged by three mechanisms: (i) not all Time-Right holders exercise physical redemption—many choose cash settlement or secondary-market transfer; (ii) physical redemptions are spread across a 36-month window, smoothing peak demand; and (iii) the safety factor $\phi = 0.80$ already embeds a 20% buffer against the cancellation rate.

However, the structural gap creates a tail risk analogous to fractional-reserve banking: if an exogenous shock (pandemic, natural disaster, regulatory action) simultaneously elevates physical redemption demand and suppresses hotel capacity, a cascade of redemption failures could trigger a market-wide confidence collapse. Mitigations include: tier-specific ω calibrated to each hotel tier's cancellation history; a mandatory reserve pool holding 5–10% of issuance proceeds as a liquidity backstop; time-window diversification requirements to prevent redemption clustering; and dynamic ω adjustment based on real-time cancellation data.

We recommend that the safety factor ϕ be treated as a tunable regulatory parameter rather than a fixed model constant. At $\phi = 1.0$ (statistically neutral), $\omega = 1.61\times$ and the required occupancy rises to 62%—equal to the expected rate, leaving no safety margin.

At $\phi = 0.60$ (highly conservative), $\omega = 0.97\times$ and the system is over-collateralized but hotel revenue is minimized. The current $\phi = 0.80$ represents a middle ground, but its appropriateness depends on the specific hotel tier and market conditions.

6.10 Reverse Stress Test: When Does the System Break?

Rather than asking "what happens under predefined stress scenarios," we invert the question: **what combination of PD and correlation causes the Senior tranche to lose investment-grade status?** We run 2,000 Monte Carlo paths across a grid of PD multipliers (1× to 12× calibrated baseline) and systematic correlation coefficients ($\rho_{\text{sys}} \in [0.70, 0.95]$). The calibrated PD baseline (pool average 9.7%) is scaled upward to simulate progressively severe credit deterioration.

Table 14: Reverse stress test: Senior EL under joint PD–ρ escalation

PD Mult.	$\rho = 0.70$	$\rho = 0.80$	$\rho = 0.90$	$\rho = 0.95$	**Break Condition**
1× (baseline)	0.07% (Aa)	0.16% (A)	0.30% (A)	0.48% (Baa)	–
2×	0.32% (Baa)	0.55% (Baa)	**1.03% (Ba)**	**1.36% (Ba)**	$\rho \geq 0.90$
3×	0.90% (Baa)	**1.33% (Ba)**	**2.05% (Ba)**	**2.51% (Ba)**	$\rho \geq 0.80$
5×	**4.27% (B)**	**4.90% (B)**	**6.01% (B)**	**6.74% (B)**	Always

The system's breaking point is clear: **when pool PD doubles to ≈19% AND correlation rises to 0.90, Senior EL crosses 1% (Ba territory).** At PD×3 with $\rho \geq 0.80$, Senior is firmly below investment grade. The Mezzanine and Junior tranches register zero losses across all scenarios because the 32% Senior layer absorbs the full pool-level default impact before losses attach to subordinate tranches—the structural protection works as designed.

6.10.1 Depression Scenario: 2009-Style Hotel Market Collapse

We simulate a severe depression scenario calibrated to the 2008–2009 financial crisis, when U.S. hotel occupancy fell 15–20 percentage points and RevPAR declined 25–30%. In this scenario: economy hotels face 60% default rates (vs. 15.7% calibrated), comfort 25%, upscale 10%, luxury 3%; systematic correlation spikes to $\rho = 0.95$; and secondary-market liquidity freezes, forcing all Time-Right holders into physical redemption at maturity.

Table 15: Depression scenario: Tranche losses under systemic hotel market collapse

Tranche	**Mean EL**	**VaR 99%**	**Status**
Senior	4.66%	82.3%	**B–Caa**: tail risk extreme
Mezzanine	0.02%	0.0%	Aaa: protected by 32% Senior layer
Junior	0.00%	0.0%	Aaa
Equity	0.00%	0.0%	Aaa

Under depression conditions, Senior expected loss rises to 4.66% (B territory), and the 99% VaR reaches 82.3%—meaning that in the worst 1% of depression paths, the Senior tranche loses over 80% of its value. The 32% Senior credit support absorbs the expected loss but offers limited protection in the extreme tail. Mezzanine and Junior tranches remain unaffected because even a 60% economy-hotel default rate, when diluted across the full pool (which includes comfort, upscale, and luxury hotels with far lower depression PDs), does not breach the Senior attachment point in expectation.

The system survives depression in expectation but not in the tail. This is the honest risk assessment: the Time-Right ABS structure is robust to severe but not catastrophic credit deterioration.

6.10.2 Physical Constraint: The True Bank Run Scenario

The over-issuance multiplier $\omega = 1.29\times$ has a hard physical limit that no diversification mechanism can circumvent. For a 60-room hotel, annual physical capacity is 21,900 room-nights. At $\omega = 1.29$, the hotel issues 28,251 Time-Rights—6,351 more than can possibly be fulfilled. In normal conditions, this gap is bridged by the 62% occupancy rate (only 13,578 room-nights are actually consumed) and settlement-mode diversification (not all holders choose physical redemption).

In a systemic crisis, this bridge collapses:

- All 28,251 holders demand physical redemption simultaneously (secondary market frozen, cash reserves distrusted).
- Physical capacity is 21,900 room-nights—**6,351 holders (22%) cannot be fulfilled under any circumstances.**
- The platform's 3% reserve pool covers only 13.3% of these unfilled claims—the system is insolvent.

This is fractional-reserve banking with a 3% reserve ratio. In 2008, banks with 10% capital requirements failed. The structural gap of 6,351 room-nights per hotel is not a parameter to be tuned—it is a physical invariant that no financial engineering can eliminate. The only genuine mitigant is that, unlike mortgage-backed securities whose underlying assets (houses) cannot be consumed, hotel room-nights have genuine demand elasticity: even in a crisis, some people need to sleep somewhere. This physical demand floor is the system's only structural advantage over pure financial assets—and it is a thin one.

6.10.3 Calibration Sensitivity: The 50% Haircut Assumption

Our calibrated PD model applies a 50% haircut to observed platform exit rates, assuming half of delistings represent true defaults. This assumption is unverified and drives the entire credit risk calibration. If the true default ratio is 25% (more optimistic), the pool PD falls to 4.9% and Senior EL drops to $\approx$0.04% (Aaa). If it is 75% (more pessimistic), the pool PD rises to 14.6% and Senior EL reaches $\approx$0.48% (Baa). The 50% assumption thus spans a $12\times$ range in Senior EL (0.04% to 0.48%)—still investment-grade across the entire plausible range, but the precise rating is indeterminate without verified default incidence data. We emphasize that the four-month observation window (March–June 2024) cannot capture seasonal cycles, macroeconomic fluctuations, or competitive dynamics that operate on longer time scales.

Table 16: Calibration sensitivity: Senior EL under alternative default-rate haircuts

True Default / Exit Ratio	Pool PD	Senior EL	Implied
25% (optimistic)	4.9%	0.04%	Aaa
50% (baseline)	9.7%	0.16%	Aa–A
75% (pessimistic)	14.6%	0.48%	Baa

The calibrated PD model is an improvement over uncalibrated structural models, but it is not a substitute for a multi-year default incidence panel.

7 Discussion: FSR Beyond Hotels

The FSR framework generalizes beyond hotel room-nights. Table 17 maps the FSR 4-tuple to four service categories:

Table 17: FSR instances across service categories

Category	Capacity (c)	Window (T)	Location (ℓ)	Type (σ)
Hotel	Room-nights	Check-in date	GPS coordinates	hotel
Flight	Seats	Departure date	Route (origin-dest)	flight
Concert	Tickets	Event date	Venue	event
Coworking	Desk-days	Calendar month	Office address	workspace

7.1 Diversification Theorem (Conjectured)

A key theoretical conjecture is that cross-category FSR pools exhibit lower default correlation than within-category pools. Hotel FSR defaults are driven by tourism seasonality; flight FSRs by business/holiday dual-peak cycles; concert FSRs by idiosyncratic event risk. These distinct business-cycle exposures suggest that multi-category FSR asset pools would achieve superior diversification to single-category geographic dispersion alone. Formal proof of this conjecture, and empirical validation with flight FSR data, is deferred to future work.

7.2 General Equilibrium Implications

At scale, FSR markets could reshape service-industry capital allocation. Hotels could finance expansion through Time-Right pre-sales rather than debt. Airlines could hedge demand risk by issuing flight FSRs. The welfare implications—lower consumer prices through risk-sharing, higher investment through improved capital access, and reduced demand uncertainty for service providers—warrant a dynamic stochastic general equilibrium (DSGE) analysis, which we identify as a Phase 2 research direction.

7.3 The Liquidity Cold-Start Problem

The continuous secondary-market offset mechanism—the system's primary user-facing innovation—depends on a liquid secondary market. At launch, liquidity is zero: there are no historical prices, no market makers beyond the platform itself, and no user confidence in the ability to exit positions. This creates a chicken-and-egg problem: without liquidity, the market-driven discount mechanism cannot function; without functioning discounts, users have no incentive to purchase Time-Rights; without demand, liquidity cannot develop.

Three structural mitigations exist but none are costless. First, the platform can act as the market maker of last resort, guaranteeing a bid at $P_{\text{issue}} \cdot (1 - f_{\text{trade}})$ until organic liquidity develops—effectively subsidizing early adopters with platform capital. Second, a liquidity mining program can reward early market participants with additional Time-Rights or fee rebates, analogous to the incentive structures that bootstrapped decentralized exchange liquidity. Third, institutional anchor investors can provide baseline demand by committing to purchase a minimum volume of Time-Rights at each issuance, creating a permanent bid that retail users can sell into.

Each of these imposes costs on the platform or requires institutional partnerships that do not yet exist. To illustrate the scale of the problem: achieving a secondary-market trading volume sufficient to provide a 4.4% effective user discount (as calibrated in Section 5.2) requires approximately 1.56 million Time-Rights to change hands annually—roughly 55% of total issuance. In month one of platform operation, trading volume is zero. The platform must either subsidize discounts directly (converting the market-driven mechanism into a platform-funded one, at least temporarily) or accept that early adopters will receive inferior economics until organic liquidity develops. Neither path is modeled in our baseline calibration. The cold-start problem is not a model parameter—it is a real-world coordination challenge. Its solution lies in market design, not in financial engineering.

7.4 Regulatory and Market Structure Considerations

The introduction of FSR as a new asset class raises regulatory questions that parallel those faced by early ABS and derivative markets.

The over-issuance mechanism creates a structural short position whose tail-risk properties are analyzed in Section 6.9. Beyond over-issuance, two additional regulatory dimensions deserve attention. First, the cross-border nature of tokenized FSRs introduces jurisdictional complexity: a Time-Right issued by a Chengdu hotel, traded on a Singapore-based secondary market, and held by a European investor falls under three regulatory regimes. The permissioned token standard (ERC-3643 or equivalent) provides an identity and compliance layer, but international coordination on FSR classification—as a security, a commodity, or a novel instrument—remains an open legal question.

Second, market microstructure considerations arise from the ternary settlement mechanism. The physical redemption option ($\alpha < 1$) creates a convenience yield for flexible travelers, analogous to the convenience yield in commodity futures markets [9]. The interaction between this convenience yield and the secondary market price premium η_t determines the equilibrium distribution of settlement choices, which in turn affects the risk profile of each ABS tranche. A structural model of this equilibrium is a natural next step.

8 Conclusion

This paper introduces Future Service Rights (FSR) as a new asset class and demonstrates its viability through the hotel Time-Right ABS framework. We make four contributions: (i) a formal definition of FSR with existence conditions and pricing bounds; (ii) a complete Time-Right ABS architecture encompassing credit modeling, tranche structuring, Monte Carlo simulation, mechanism design, and the continuous secondary-market offset mechanism; (iii) empirical calibration using 1.7 million hotel price observations from Chengdu; and (iv) a complete money-flow stakeholder analysis demonstrating ¥1,227 of net value creation per Time-Right and a 55% hotel revenue advantage per available room-night over OTA distribution.

The Time-Right model delivers 56% higher annual hotel revenue than OTA distribution (¥35.2M vs. ¥22.6M per hotel-year), driven by the over-issuance multiplier and occupancy-independent revenue certainty. The continuous secondary-market offset mechanism transforms Time-Rights into a market-driven voucher system, where user discounts are funded by secondary-market liquidity rather than hotel margin compression. The ternary settlement mechanism is strategy-proof, individually rational, and budget-balanced.

Several important limitations bound our conclusions. First, our empirical evidence is limited to a single city (Chengdu) and a single service category (hotels). More critically, the four-month observation window (March–June 2024) cannot capture seasonal cycles, macroeconomic fluctuations, or competitive dynamics that operate on annual or multi-year scales. This temporal limitation is more fundamental than the geographic one: expanding to 50 cities would not compensate for a data window too short to observe a full credit cycle. Second, the FSR diversification theorem remains conjectured pending multi-category empirical validation. Third, while we have resolved the structural-model divergence by constructing a data-calibrated PD model from hotel survival data, the calibration relies on the assumption that approximately 50% of platform exits represent true defaults—a reasonable but unverified premise. Direct hotel default incidence data would further tighten the calibration. Fourth, while the per-night Time-Right revenue (¥1,054) is comparable to OTA-net revenue (¥1,032), the over-issuance multiplier ($\omega = 1.29\times$) and occupancy-independent certainty combine to deliver 56% higher annual hotel revenue (¥35.2M vs. ¥22.6M). The economic value of Time-Rights lies primarily in this volume-certainty combination, not in per-night price improvement. Fifth, the continuous secondary-market offset mechanism—the system's primary user-facing innovation—depends on liquid secondary markets that do not exist at launch. The cold-start problem (Section 7.3) is a market-design challenge that no amount of Monte Carlo simulation can resolve; it must be addressed through platform subsidies, liquidity mining, or institutional anchor investors, each of which imposes costs not modeled in our baseline calibration.

Future work should prioritize four directions, in order of urgency. First and most critically, the 50% haircut assumption—that half of observed platform exits represent true hotel defaults—must be validated or replaced with direct hotel default incidence data. This single unverified premise drives the entire credit risk calibration, and until it is empirically anchored, all tranche safety conclusions carry an irreducible model risk. Second, multi-category empirical validation with flight, event, and coworking FSR data is needed to test the diversification conjecture. Third, the copula sensitivity framework must be integrated into the full ternary-settlement engine to produce unified risk estimates that capture both settlement-mechanism loss absorption and tail dependence simultaneously.

Fourth, live market deployment with on-chain settlement and real user behavior data would resolve the liquidity cold-start problem identified in Section 7.3.

The FSR framework is not merely a paper about hotel securitization. It is the foundation of a research program on generalized service-capacity financialization.

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Appendix: Parameter Traceability

All model parameters are classified by evidentiary status. **CALIBRATED**: derived from external data or published literature. **ASSUMPTION**: reasonable estimate with literature support, but not directly calibrated to this dataset. **JUDGMENT**: author judgment; should be subjected to sensitivity analysis. **DERIVED**: computed from other parameters.

Table 18: Parameter traceability: key model inputs and their sources

Parameter	**Value**	**Status**	**Source / Sensitivity**
Credit Risk			
PD cap	0.50	JUDGMENT	See Table 11 for uncapped
PD floor	0.001	JUDGMENT	Minimal impact (few hotels near floor)
Volatility floor	0.05	JUDGMENT	See Section 4 for level multipliers
DD floor	0.10	JUDGMENT	Prevents DD $\to$ 0 instability
Default barrier	$0.55V_i$	JUDGMENT	Sensitivity: 0.45–0.65 tested
PD calibration	2.50	ASSUMPTION	Bharath & Shumway (2008): 1.5–3.0 range
GARCH (ω, α, β)	(1e-5, 0.10, 0.85)	ASSUMPTION	Standard financial time-series defaults
ρ_{sys}	0.70	JUDGMENT	Sensitivity: 0.50–0.95 (Section 6.5)
Pricing & Issuance			
Discount rate r	0.08	JUDGMENT	Sensitivity: 5–12% (Section 6.6)
Convergence λ	0.80	JUDGMENT	Price decay speed; not yet sensitivity-tested
d_{issue}	0.10	JUDGMENT	Sensitivity: 0–30%
Platform markup m	0.08	JUDGMENT	Sensitivity not yet tested
α (physical)	0.70	JUDGMENT	$\alpha \geq \delta(T)(1-d)$ for IR
ϕ (safety factor)	0.80	JUDGMENT	Sensitivity: 0.60–1.00 (Section 6.9)
ω	1.29	DERIVED	$\omega = (1/\rho) \cdot \phi$
Calibrated PD Model			
Default / exit ratio	0.50	JUDGMENT	Sensitivity: 0.25–0.75 (Table 16)
Attrition rates	Tier-specific	CALIBRATED	14,851 hotels, Mar–Jun 2024
Structuring			
Tranche ratios	68/20/8/4%	JUDGMENT	Industry-standard for CMBS
Coupons	4.5/6.5/9.5%	JUDGMENT	Aligned with target ratings
Reserve pct	3.0%	JUDGMENT	Sensitivity not yet tested

The complete parameter audit, including 200+ parameters across 8 engine modules, is maintained in `src/model_params.py` in the replication repository.

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